Your Name Date

C Level Work

Lab Problem 1:

Objective: Create a program that will have delay steps from 500 to 2500 in 500 ms steps and step down from 2500 to 500 in 500 ms steps.

Code: (Note, no opening comment is a negative)

```
const int LED=10;           //define LED for Pin 9
void setup()
{
  pinMode (LED, OUTPUT); //Set the LED pin as an output
}
void loop()
{
  for (int i=500; i<=2500; i=i+500)
  {
    digitalWrite(LED, HIGH);
    delay(i);
    digitalWrite(LED, LOW);
    delay(i);
  }

  for (int i=2500; i>=500; i=i-500)
  {
    digitalWrite(LED, HIGH);
    delay(i);
    digitalWrite(LED, LOW);
    delay(i);
  }

}
```

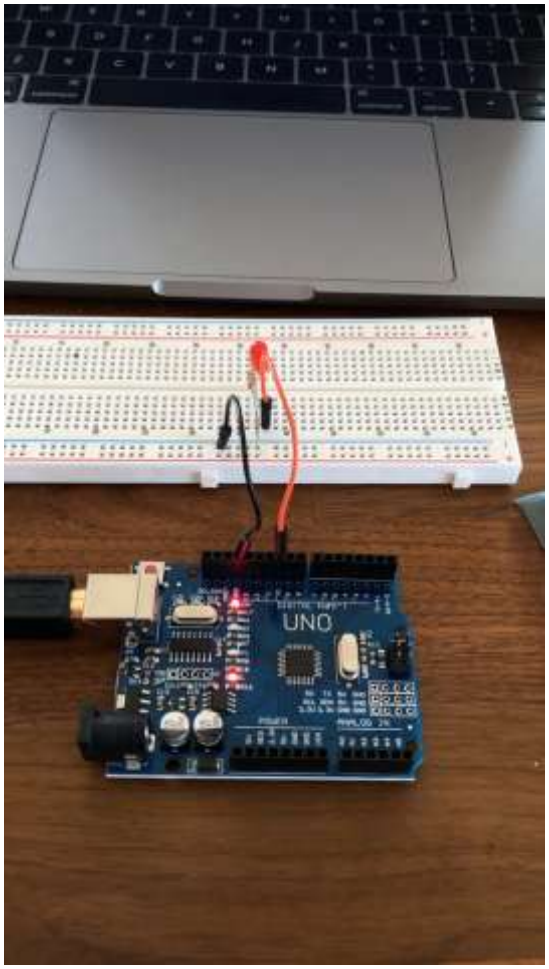
No extra commentary

Link/Video: https://youtu.be/HHHHHHHHHH

Your Name Date

B Level Work

Lab problem 1: Post a video of your LED turning on and off in your lab report.

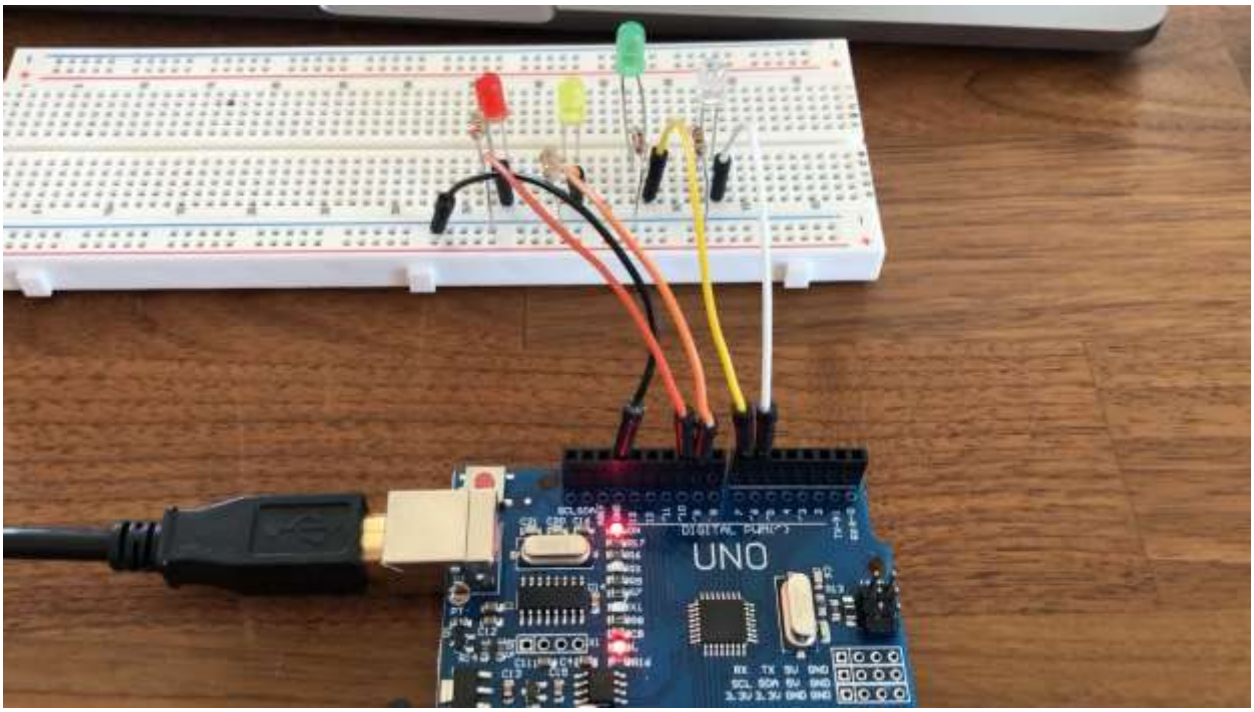


Some extra commentary

Lab Problem 2: Post in the lab report a second video with these modifications as well as comments on what the pattern it is doing.

I modified my 4 LEDs to match the pattern of the lights shown on the Knight Rider TV series Youtube video. My loop pattern looks like the following at 50 ms: pin 10 -> pin 9 -> pin 7-> pin 6 -> pin 7 -> pin 9 (Loop starts cycle again at Pin 10 after this)

Your Name Date



ARDUINO CODE: **Code:** (Note, no opening comment is a negative)

```
void setup() {
pinMode(10,OUTPUT);  //Initialize Pin
pinMode(9,OUTPUT);  //Initialize Pin
pinMode(7,OUTPUT);  //Initialize Pin
pinMode(6,OUTPUT);  //Initialize Pin
}
void loop() {
digitalWrite(10,HIGH); //Set the LED On
delay(50);          //Wait for 50 ms
digitalWrite(10,LOW);  //Set the LED Off
delay(50);          //Wait for 50 ms

digitalWrite(9,HIGH); //Set the LED On
delay(50);          //Wait for 50 ms
digitalWrite(9,LOW);  //Set the LED Off
delay(50);          //Wait for 50 ms

digitalWrite(7,HIGH); //Set the LED On
delay(50);          //Wait for 50 ms
digitalWrite(7,LOW);  //Set the LED Off
delay(50);          //Wait for 50 ms

digitalWrite(6,HIGH); //Set the LED On
delay(50);          //Wait for 50 ms
digitalWrite(6,LOW);  //Set the LED Off
delay(50);          //Wait for 50 ms

digitalWrite(7,HIGH); //Set the LED On
delay(50);          //Wait for 50 ms
digitalWrite(7,LOW);  //Set the LED Off
```

Your Name Date

```
delay(50);          //Wait for 50 ms

digitalWrite(9,HIGH); //Set the LED On
delay(50);          //Wait for 50 ms
digitalWrite(9,LOW);  //Set the LED Off
delay(50);          //Wait for 50 ms


}
```

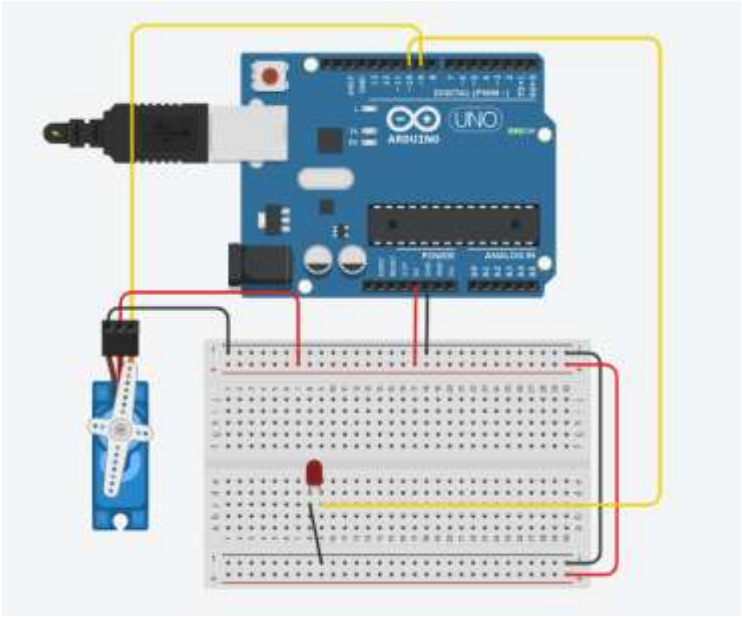
Your Name Date

A Level Work

Problem 2:

- **Objective:**
  Connect a servo with a directional arm according to the directions in the previous servo lab (you can likely leave out the regulator and hook the servo up to the Arduino 5-volt supply). Allow the user to enter an angle and have the servo turn to that angle. If the angle entered is invalid (outside of 0-180°) a red light should light on the breadboard. Post your code and a video to the class blog.

- **Materials:**
  Servo, red led
- **Setup:**
  I used the previous two lab examples, the one using 1s and 0s to turn on and off a blue led, and the one where rgb values for a rgb led are declared in the serial monitor. However, I encountered a problem where the values I input where incorrect and would not move the servo due to being outside the range of 0 and 180. I figured it was a problem with datatypes between character and integers and tried setting my value between two ''. However, I eventually realized that the int I set to hold my serial monitor input was actually set as char instead of int.



I would prefer you upload videos as a separate file along with the lab document. I cannot run embedded videos within the Canvas shell.

Servo Serial Input.mp4

- **Answer:**

```
//Program moves servo based on serial input value
//Programmers name was here.  Date would be a good addition.
//Define LED pins
#include <Servo.h>

const int RED   =10;
const int SERVO =9;
int val;
Servo myServo;
```

Your Name Date

```
void setup()
{
  Serial.begin(9600); //Serial Port at 9600 baud

  //Set pins as outputs
  pinMode(RED, OUTPUT);
  myServo.attach(SERVO);
  Serial.print("Input desired servo angle\n");
}

void loop()
{
  //Keep working as long as data is in the buffer
  while (Serial.available() > 0)
{
    val = Serial.parseInt();  //Value set to integer
    //Serial.println(val); //debugging
    if ((val > 180) || (val < 0))
     digitalWrite(RED, HIGH);
     else
      digitalWrite(RED, LOW);
      if(val<0) //When value is below 0, entered value and error message printed
      {
      Serial.print("Value entered: ");
      Serial.println(val);
      Serial.print("Value entered cannot be below 0\n");
      }
      else if(val>180) //When value is above 180,entered value and error message
printed
      {
      Serial.print("Value entered: ");
      Serial.println(val);
      Serial.print("Value entered cannot be above 180\n");
      }

//Move Servo
    if ((Serial.read() == '\n') && ( (val>=0)&&(val<=180) )) //Move Servo
condition set so it only moves within certain values
    {
      //set SERVO
      myServo.write(val);
      Serial.print("Servo moved to ");
      Serial.println(val);
      delay(15);
    }
}
}
```